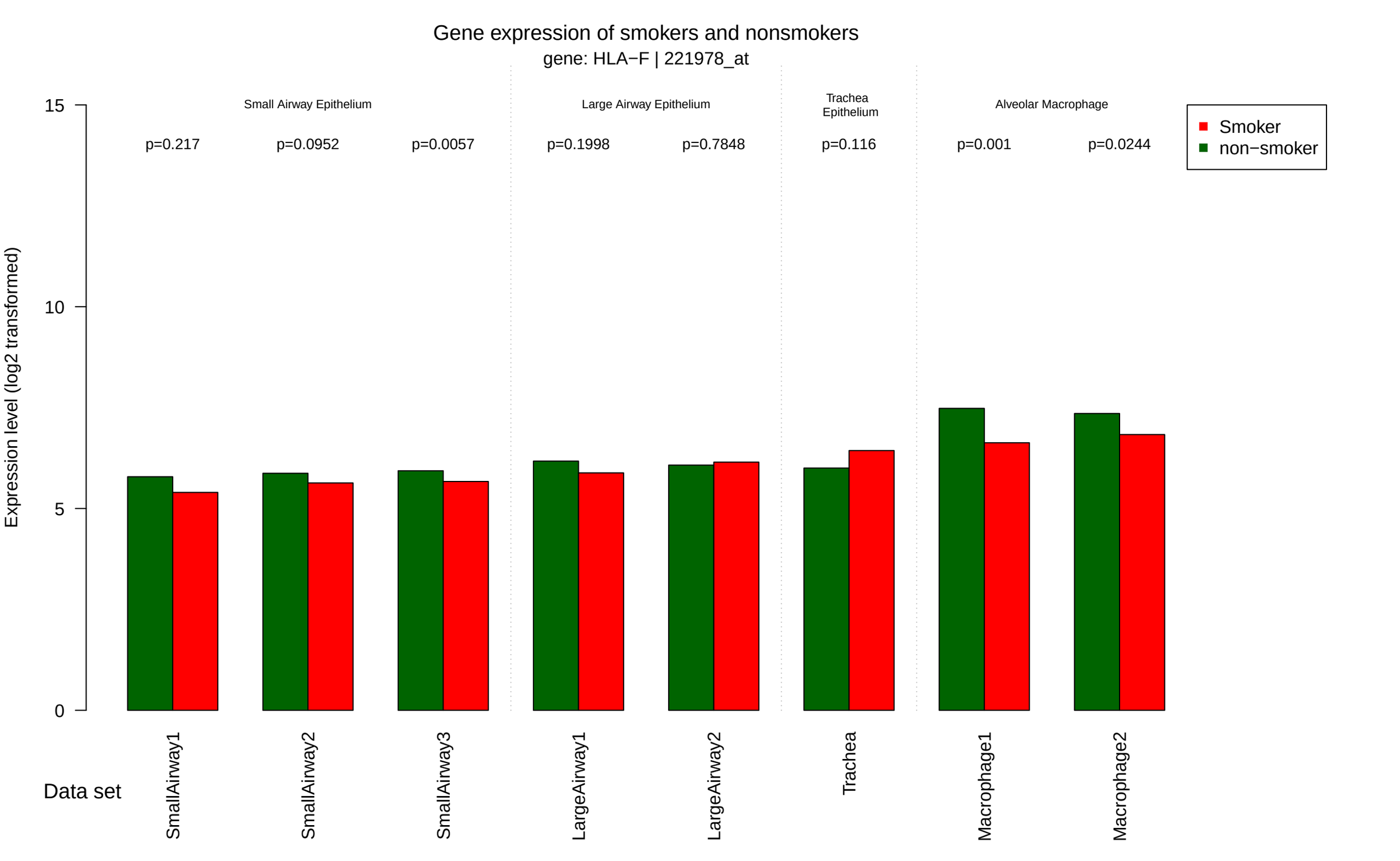
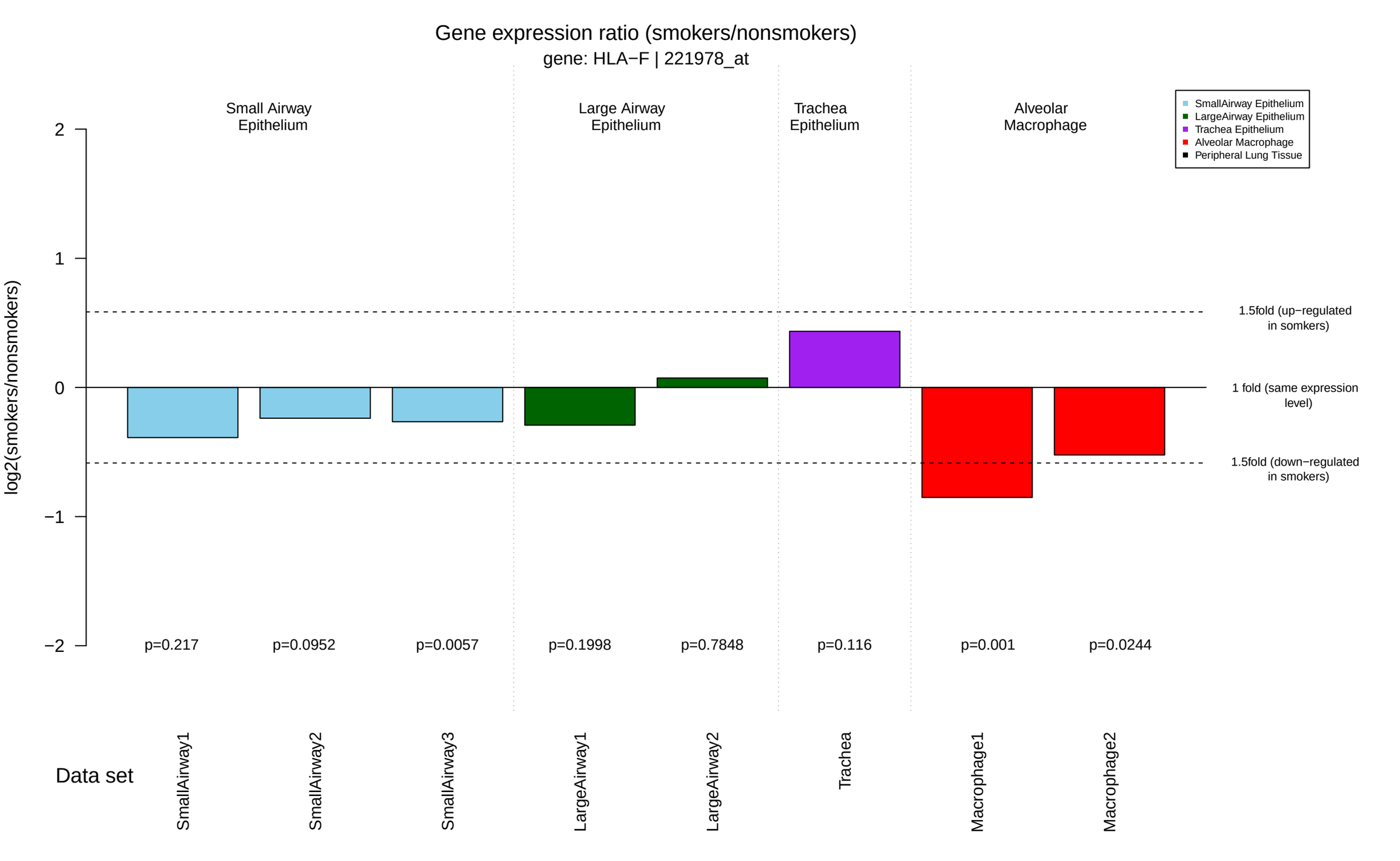
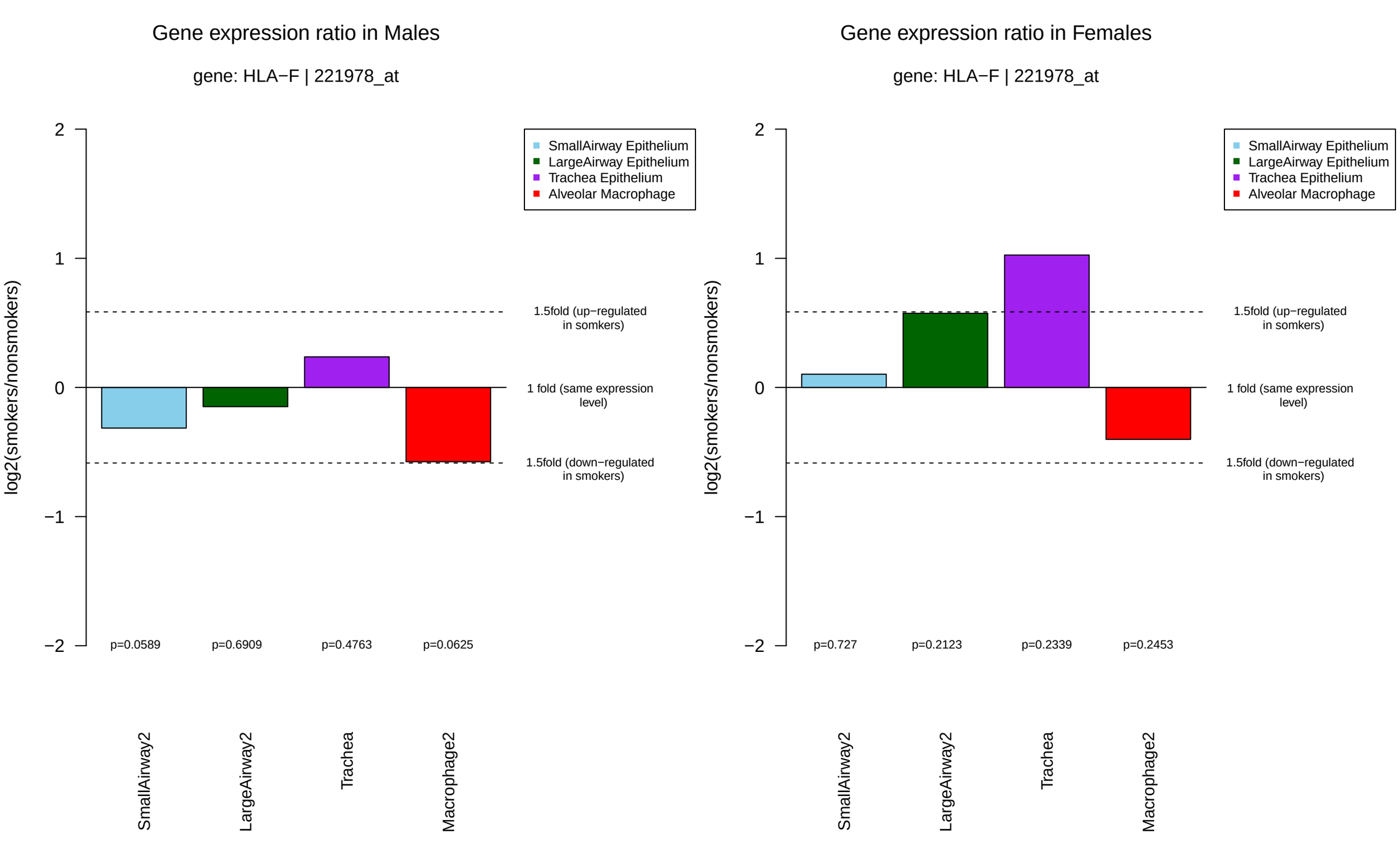
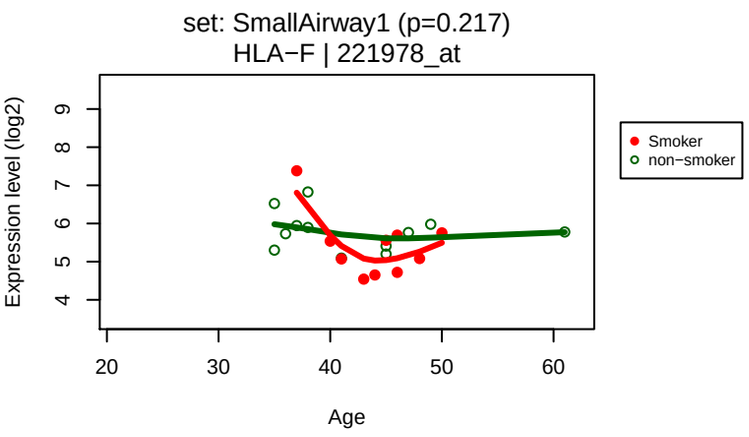
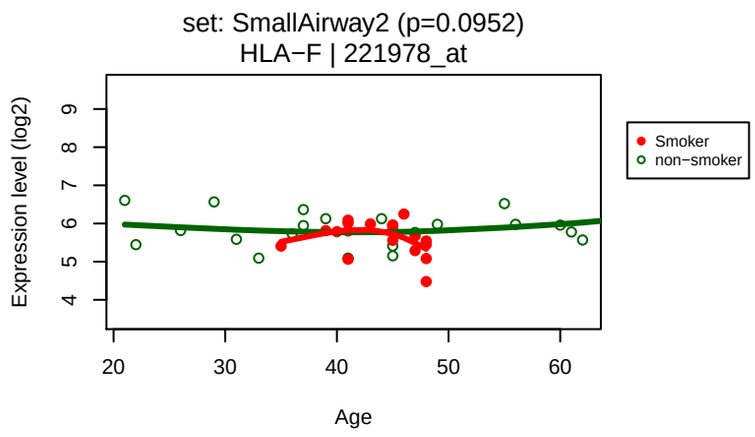
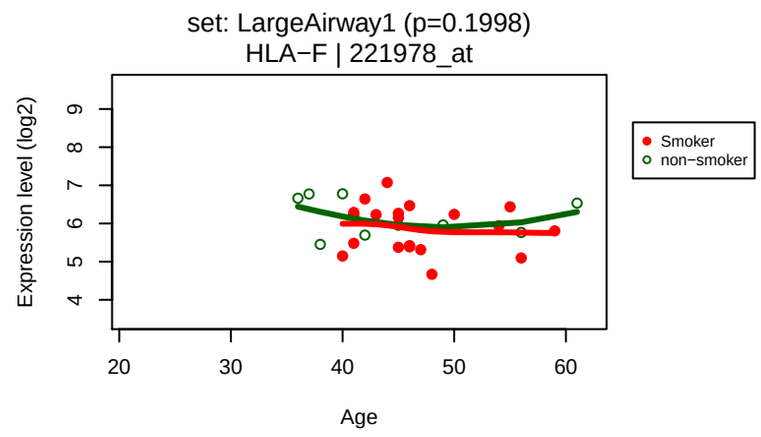

Gene expression of smokers and nonsmokers
gene: HLA−F | 221978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.217
p=0.0952
p=0.0057
p=0.1998
p=0.7848
p=0.116
p=0.001
p=0.0244
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

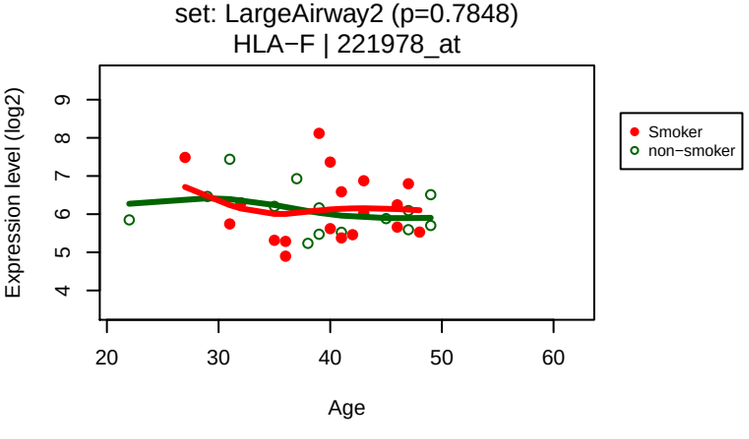

Gene expression ratio (smokers/nonsmokers)
gene: HLA−F | 221978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.217
p=0.0952
p=0.0057
p=0.1998
p=0.7848
p=0.116
p=0.001
p=0.0244
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

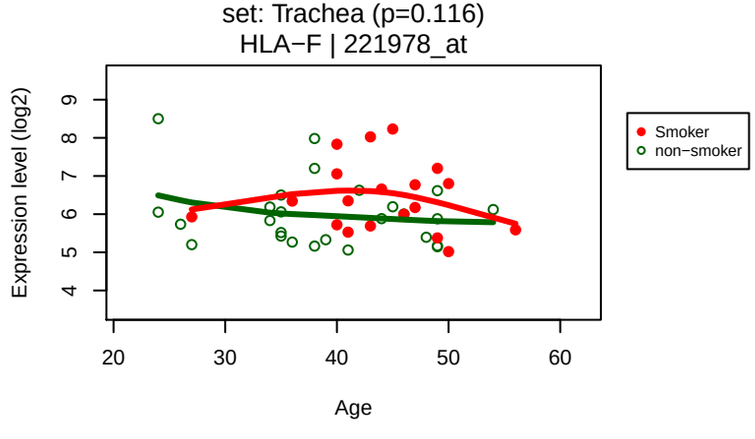

Gene expression ratio in Males
gene: HLA−F | 221978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0589
p=0.6909
p=0.4763
p=0.0625
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HLA−F | 221978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.727
p=0.2123
p=0.2339
p=0.2453
SmallAirway2
LargeAirway2
Trachea
Macrophage2

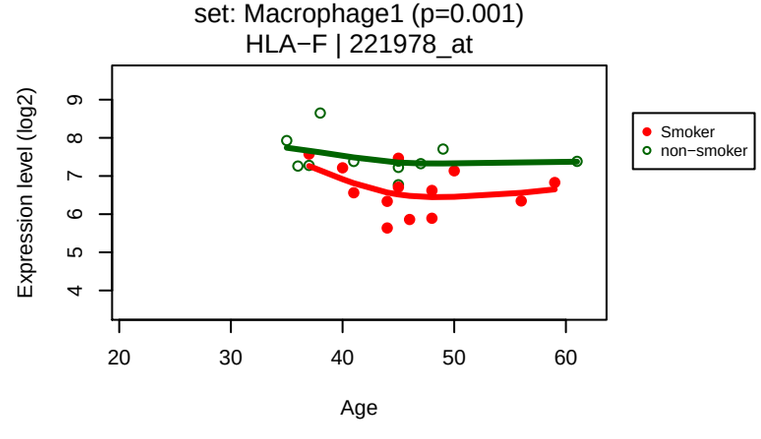

set: SmallAirway1 (p=0.217)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker

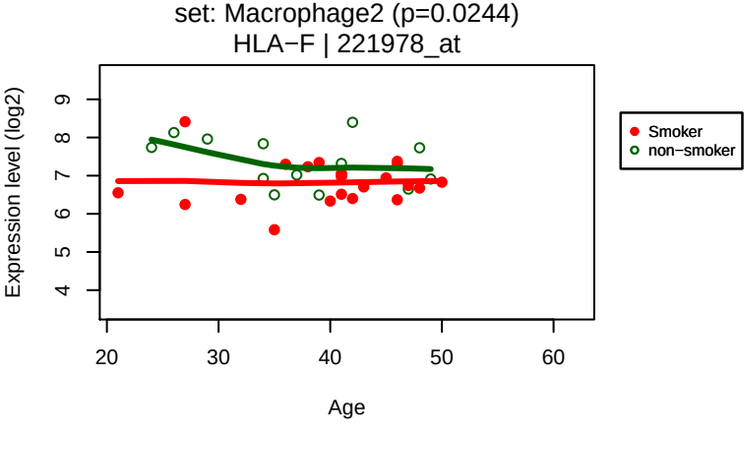

set: SmallAirway2 (p=0.0952)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker

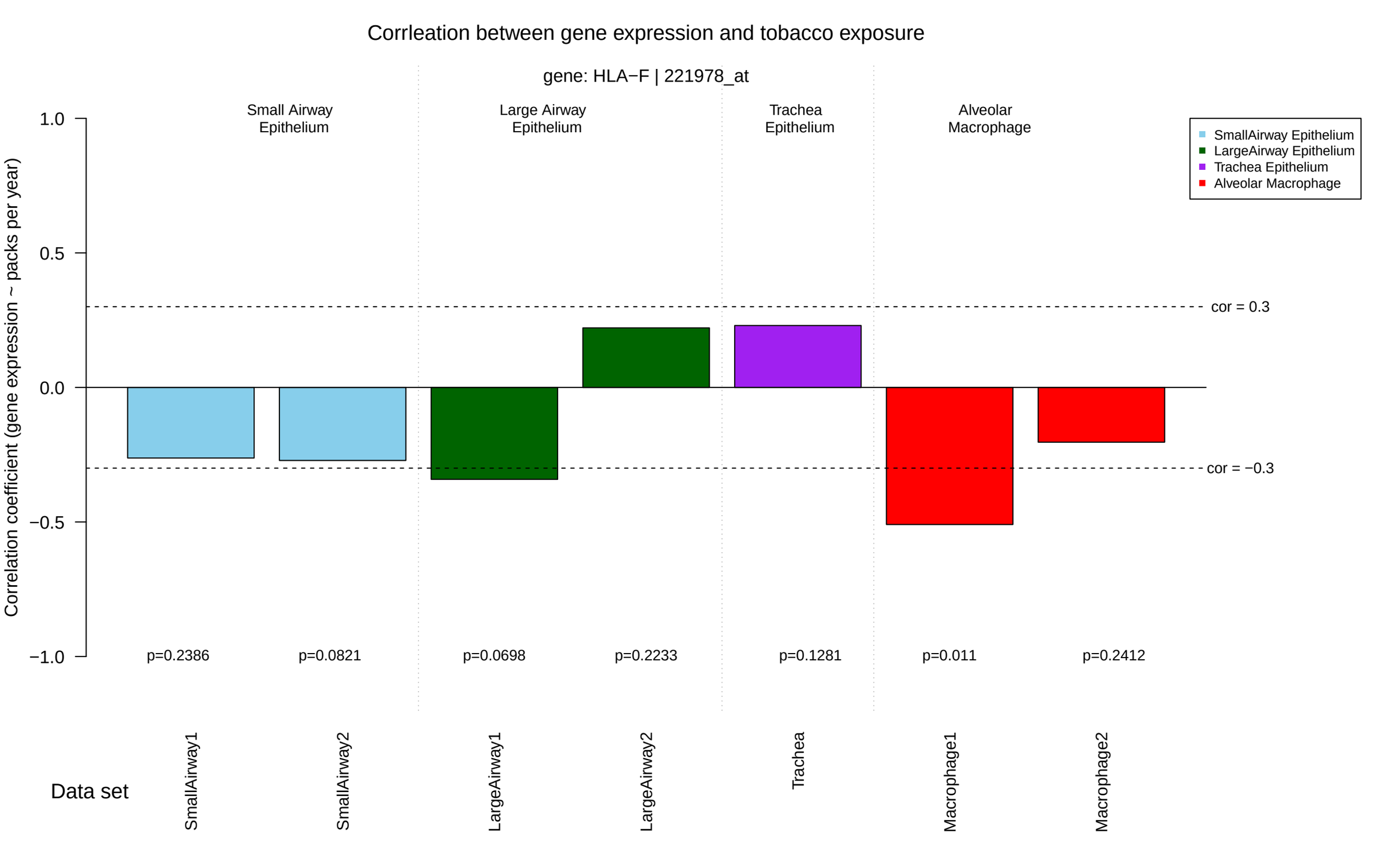

set: LargeAirway1 (p=0.1998)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7848)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.116)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.001)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0244)
HLA−F | 221978_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HLA−F | 221978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2386
p=0.0821
p=0.0698
p=0.2233
p=0.1281
p=0.011
p=0.2412
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2